

Indian Equity Technical Advisory Agent & Swing Screener

Mathematical Reference Manual, Strategy Specifications & Algorithmic Audit

Target Horizon: Swing (3-10 Day Hold)	Document Version: 2.0.0 (Production)	Asset Universe: NSE Nifty 100 / Nifty 50
Execution Type: Non-Custodial Decision Support	Date: September 2026	Status: Validated Live on Kite Connect

1. Executive Summary & System Philosophy

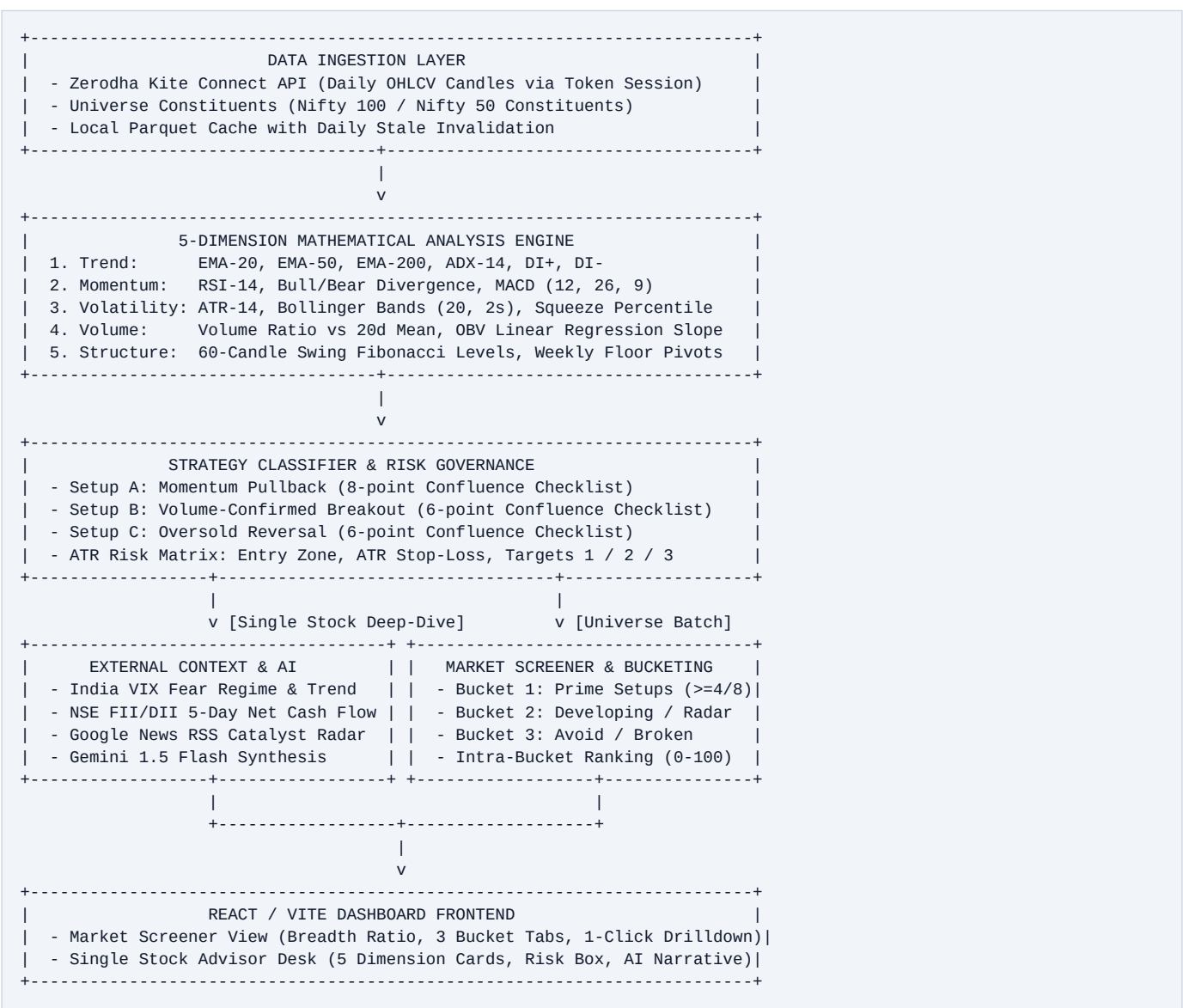
The Advisory System is engineered to address the twin vulnerabilities that undermine retail swing traders: **(1) The Discovery Dilemma** (missing high-probability setups developing at structural inflection zones due to the impossibility of manually auditing 100+ stock charts daily), and **(2) The Psychological Trap** (chasing overbought momentum, entering at extreme extensions, and trading without deterministic, mathematically bounded risk levels).

OPERATIONAL MANDATE & RISK GOVERNANCE

Core Architectural Axioms:

- Strictly Non-Custodial:** Zero automated order placement. The engine operates purely as an institutional-grade intelligence and risk-governance advisory terminal.
- Strict Separation of Math vs. Generative AI:** All trend states, momentum indicators, confluence tallies, stop-losses, and profit targets are strictly derived via *deterministic Python algorithms*. The Gemini 1.5 Flash LLM acts solely as a natural-language synthesis layer and is strictly prohibited from recalculating or altering any mathematical metric.
- Daily Swing Cadence:** Built upon Daily (EOD) OHLCV candles calibrated for 3-to-10-day holding periods, insulating the trader from intraday noise.

2. End-to-End System Architecture



SYSTEM PIPELINE & DATA FLOW

Architectural Pipeline Summary:

- 1. **Data Layer:** Ingests historical daily candles directly from Zerodha Kite Connect with intelligent disk caching.
- 2. **Analytics Engine:** Computes 5 mathematical dimensions with zero forward-looking bias.
- 3. **Strategy & Risk Layer:** Evaluates 8-point and 6-point confluence criteria and produces strict ATR price envelopes.
- 4. **Distribution Layer:** Feeds both the single-stock deep diagnostic and the market-wide 3-tier screener simultaneously.

3. Dimension 1: Trend Architecture (EMA Stack & ADX-14)

3.1 Exponential Moving Averages (EMA):

$$EMA_t = (Close_t \times \alpha) + (EMA_{t-1} \times (1 - \alpha))$$

Where $\alpha_{20} = 2/21 \approx 0.0952$ (Fast Trend) | $\alpha_{50} = 2/51 \approx 0.0392$ (Intermediate Trend) | $\alpha_{200} = 2/201 \approx 0.00995$ (Macro Trend)

3.2 Trend Classification State Machine:

Trend State	Mathematical Criterion	Swing Trading Actionability
STRONG_BULL	Price > EMA ₂₀ > EMA ₅₀ > EMA ₂₀₀	All institutional horizons bullish. Prime candidate for Momentum Pullback.
BULL	Price > EMA ₅₀ and EMA ₂₀ > EMA ₅₀	Confirmed intermediate uptrend. Dips toward EMA ₂₀ offer high R:R.
NEUTRAL	Price between EMA ₂₀ and EMA ₅₀ , or EMAs entwined	Consolidation or transition. Wait for clear expansion before allocating capital.
BEAR	Price < EMA ₅₀	Downward momentum dominant. Disqualified from long swing positions.
STRONG_BEAR	Price < EMA ₂₀₀ and EMA ₂₀ < EMA ₅₀	Macro institutional distribution. Strictly classified into Bucket 3 (Avoid).

3.3 Average Directional Index (ADX-14):

$$DX = (|DI^+ - DI^-| / (DI^+ + DI^-)) \times 100 \implies ADX_{14} = WilderSmooth(DX, 14)$$

Where $TR = \max(H - L, |H - C_{prev}|, |L - C_{prev}|)$ and Directional Movement DM+, DM- are 14-period smoothed.

ADX-14 Range	Regime Identifier	Implication for Swing Traders
ADX < 20	RANGING / CHOPPY	Trendless consolidation. High false-breakout rate. Disqualifies trend strategies.
20 <= ADX < 25	DEVELOPING	Trend initiation phase. Early swing entry window.
25 <= ADX <= 40	STRONG TREND	Optimal swing trading environment. Highest statistical follow-through.
ADX > 40	VERY STRONG / CLIMACTIC	Mature trend extension. Trailing stops must be tightened; reversal risk elevated.

4. Dimension 2: Momentum & Divergence Dynamics (RSI & MACD)

4.1 Relative Strength Index (RSI-14):

$$RSI = 100 - (100 / (1 + RS)), \text{ where } RS = AvgGain_{14} / AvgLoss_{14}$$

14-Period Exponential Smoothed Momentum Oscillator (Range: 0 to 100)

RSI Range	Zone Classification	Strategic Utility
RSI > 70	OVERBOUGHT	Momentum overextended. New swing longs strictly prohibited. Wait for cooling.
55 < RSI <= 70	MOMENTUM_ZONE	Bullish thrust. Favorable for trailing runners; not optimal for fresh entry.
38 <= RSI <= 58	PULLBACK_ZONE	The Primary Swing Entry Sweet-Spot. Momentum has reset to healthy equilibrium.
30 <= RSI < 38	WEAK	Selling pressure dominant. Requires structural stabilization prior to entry.
RSI < 30	OVERSOLD	Capitulation extreme. Valid exclusively for Setup C (Oversold Reversal) at deep support.

4.2 Automated Divergence & MACD (12, 26, 9):

- **Bullish Divergence:** Price_{current} ≤ min(Price₂₀)×1.01 AND RSI_{current} > min(RSI₂₀)×1.05. Selling velocity dried up.
- **MACD Crossover:** MACD Line = EMA₁₂(Close) - EMA₂₆(Close). Crossover above Signal line within 5 bars = CROSSOVER_BULLISH.

5. Dimension 3: Volatility Cycles (ATR & Bollinger Squeeze)

5.1 Average True Range (ATR-14):

$$ATR_{14} = WilderSmooth(TR, 14) \mid ATR \% = (ATR_{14} / Price) \times 100$$

Dynamic volatility yardstick for risk management. Accounts for asset-specific price dispersion.

5.2 Bollinger Bands (20, 2σ) & Volatility Squeeze Percentile:

$$BandWidth \% = ((UpperBand - LowerBand) / MiddleBand) \times 100 \ [Upper/Lower = SMA_{20} \pm 2\sigma]$$

Squeeze Percentile = Percentage of past 252 trading sessions with BandWidth narrower than current

Squeeze Percentile	Volatility State	Execution Significance
<= 20th Percentile	SQUEEZE (Coiled)	Extreme compression. Energy is stored; high-probability precursor to explosive breakout.
>= 80th Percentile + Rising ATR	EXPANSION	Breakout released. Trend acceleration in progress; risk of late-stage exhaustion.
20th to 80th Percentile	NORMAL	Standard volatility behavior. Favorable for structured swing pullbacks.

6. Dimension 4: Volume & Institutional Participation

6.1 Volume Ratio vs. 20-Day Trailing Average:

$$Volume\ Ratio = Volume_{today} / ((1 / 20) \times \sum_{i=1}^{20} Volume_{t-i})$$

Volume validates price conviction. Low volume on pullbacks indicates an absence of institutional selling.

Volume Ratio	Classification	Swing Trading Interpretation
>= 2.0x	SURGING	Institutional block accumulation. Mandatory for Setup B (Breakout) confirmation.
1.3x to 2.0x	ABOVE_AVERAGE	Healthy institutional sponsorship supporting the session.
0.8x to 1.3x	NORMAL	Standard baseline liquidity.
0.5x to 0.85x	CONTRACTING	Optimal for Pullbacks (Setup A). Confirms sellers are exhausted; no dumping.
< 0.5x	VERY_LOW	Lack of market participation. Reversals from this level lack follow-through.

6.2 On-Balance Volume (OBV) Linear Regression Slope:

$$Normalized\ Slope = Slope_{10}(OBV) / Mean(|OBV|) \ [Thresholds: > +0.003 = UPTREND, < -0.003 = DOWNTREND]$$

Disqualifies long setups when OBV is in an active DOWNTREND (indicating institutional distribution).

7. Dimension 5: Structural Geometry & Levels

7.1 60-Candle Swing Fibonacci Retracements:

The engine establishes rolling swing boundaries over the trailing 60 trading sessions (~3 months):
 $\Delta(\text{Range}) = \text{Swing High}_{60} - \text{Swing Low}_{60}$. Levels projected downward: **23.6%**, **38.2%**, **50.0%** (Gann/Dow equilibrium), **61.8%** (Golden Ratio), and **78.6%** (Deep defense).
Proximity Trigger: Price is flagged at structural support when $|\text{Price} - \text{Fib}_{\text{level}}| / \text{Price} \leq 1.5\%$.

7.2 Weekly Floor Pivot Points:

$$PP = (H + L + C) / 3 \mid R_1 = 2 \times PP - L \mid S_1 = 2 \times PP - H \mid R_2 = PP + (H - L) \mid S_2 = PP - (H - L)$$

Derived from the completed prior weekly candle. Price > PP denotes a bullish weekly institutional bias.

8. Strategy Classification & Confluence Engine

The system scores every stock against three institutional swing trading archetypes:

Strategy Archetype	Key Confluence Checklist	Qualification Bar
Setup A: Momentum Pullback (Core Swing Play)	1. $EMA_{20} > EMA_{50}$ (Mandatory) 2. Price within $\pm 1.5\%$ of EMA_{20} or EMA_{50} 3. $ADX_{14} \geq 20$ (Trending market) 4. RSI in cooling zone: $38 \leq RSI \leq 58$ (Mandatory) 5. Volume contracting ($< 0.90 \times 20d$ avg) 6. OBV trend is UPTREND (Smart money accumulating) 7. Price at Fibonacci 38.2%, 50%, or 61.8% level ($\leq 2\%$) 8. Price above Weekly Pivot Point (PP)	Min. 4 of 8 Criteria (Must satisfy #1 and #4)
Setup B: Volume-Confirmed Breakout	1. Bollinger Squeeze active (Percentile $< 30\%$) (Mandatory) 2. Volume surging $\geq 1.5 \times 20d$ avg (Mandatory) 3. OBV trend is UPTREND 4. $ADX < 45$ (Emerging trend, not exhausted) 5. $RSI < 70$ (Room to expand) 6. Price above Weekly Pivot Point (PP)	Min. 4 of 6 Criteria (Must satisfy #1 and #2)
Setup C: Oversold Reversal	1. $RSI_{14} < 35$ (Extreme oversold) (Mandatory) 2. Bullish RSI Divergence confirmed (Mandatory) 3. Price at deep Fib (61.8% or 78.6%) 4. Price at/below weekly S_1 or S_2 pivot 5. Volume expansion $\geq 1.3 \times$ on turnaround candle 6. OBV not in downward breakdown	Min. 4 of 6 Criteria (Must satisfy #1 and #2)

9. Deterministic Risk Management & Asymmetric Targets

No human or LLM opinion can override risk calculations. Stop-losses and targets are strictly bounded by volatility:

$$Stop-Loss = LTP - (M_{SL} \times ATR_{14}) \mid Risk\ Per\ Share = LTP - Stop-Loss$$

Where $M_{SL} = 2.0$ for Pullback (Setup A) | $M_{SL} = 1.5$ for Breakout (Setup B) | $M_{SL} = 1.5$ for Reversal (Setup C)

Execution Parameter	Mathematical Formulation	Risk-to-Reward Ratio (R:R)
Entry Zone	[LTP × 0.998, LTP × 1.002]	Current baseline price buffer (±0.2%)
Stop-Loss	$LTP - (M_{SL} \times ATR_{14})$	1.0R (Risk Unit)
Target 1 (Conservative)	$LTP + (1.5 \times Risk\ Per\ Share)$	1 : 1.5 (Take 50% profit; move SL to breakeven)
Target 2 (Base Case)	$LTP + (2.5 \times Risk\ Per\ Share)$	1 : 2.5 (Take 30% profit; trail remaining)
Target 3 (Runner)	$LTP + (4.0 \times Risk\ Per\ Share)$	1 : 4.0 (Full exit or trailing EMA_{20})

CAPITAL PRESERVATION GUARANTEE

Institutional 1% Risk Position Sizing Rule:

Position Quantity = floor((Total Capital × 0.01) / Risk Per Share)

Example: Account = Rs. 5,00,000 | 1% Risk = Rs. 5,000 | LTP = Rs. 2,500 | Stop-Loss = Rs. 2,420 (Risk = Rs. 80).

Position Size = floor(5,000 / 80) = **62 Shares** (Total Allocation: Rs. 1,55,000). Max loss on stop hit is exactly Rs. 4,960 (≤ 1%).

10. Market-Wide Screener & Automated 3-Tier Bucketing

To eliminate information overload, the screener parses the full universe and automatically partitions stocks into **three mutually exclusive buckets**:

Bucket	Strict Qualification Criteria	User Interface & Action Protocol
Bucket 1: PRIME SETUPS (Actionable Now)	- Signal == BUY - Confluence Count ≥ 4 / 8 - TrendState != BEAR and != STRONG_BEAR - Defined ATR Risk Matrix with R:R ≥ 1:1.5	Actionable candidates (3 to 7 stocks). Displays Entry Zone, SL, T1/T2/T3, Rank Score, and 1-Click Deep Diagnostic drilldown.
Bucket 2: DEVELOPING (On Radar / Watchlist)	- TrendState == STRONG_BULL or BULL (Price > EMA_{50}) - Setup NOT yet triggered (Extended >1.5% above EMA_{20} or RSI cooling >58) - Structure is sound, but entry is premature	Watchlist candidates (10 to 20 stocks). Generates explicit tactical advice: 'Extended +4.8% above EMA_{20} . Wait for retrace to Rs. 2,450.'
Bucket 3: AVOID / STAY AWAY (Capital Preservation)	- TrendState == BEAR or STRONG_BEAR (Price < EMA_{50} or < EMA_{200}) - Choppy consolidation (ADX < 18) or broken momentum - Institutional distribution (OBV downtrend)	The remaining market (Disqualified). Protects the trader from catching falling knives or holding dead money.

11. Multi-Factor Intra-Bucket Ranking Algorithm

Inside **Bucket 1 (Prime Setups)**, candidates are sorted from highest to lowest conviction using a multi-factor composite formula yielding a score from 0 to 100:

$$Rank\ Score = (0.35 \times S_{Confluence}) + (0.25 \times S_{Proximity}) + (0.20 \times S_{RR}) + (0.20 \times S_{Volume})$$

Composite intra-bucket score weighting structural strength, support proximity, payoff asymmetry, and volume conviction.

Score Component	Normalized Mathematical Definition	Strategic Objective
S_Confluence (35%)	$S_{Confluence} = (Confluence\ Count / Max\ Possible) \times 100$	Rewards setups satisfying 6/8 or 7/8 institutional conditions.
S_Proximity (25%)	$S_{Proximity} = \max(0, 100 - (Distance\ \% \ to\ EMA_{20} \times 25))$	Rewards stocks sitting directly at support (0.2% away = 95 pts).
S_RR (20%)	$S_{RR} = \min(100, (Target\ 2\ R:R / 2.5) \times 100)$	Rewards asymmetric payoff structures.
S_Volume (20%)	$S_{Volume} = \min(100, \max(20, Volume\ Ratio \times 50))$	Rewards institutional volume confirmation.

Intra-Bucket Sorting for Watchlist and Avoid:

- **Bucket 2 (Developing):** Sorted by $|Proximity\ \% \ to\ EMA_{20}|$ ascending. The stock closest to triggering its pullback ranks #1.
- **Bucket 3 (Avoid):** Sorted by ascending RSI (most technically broken stocks appear first).

12. External Market Context Integration

12.1 India VIX (The Fear Gauge):

Live India VIX is fetched via Zerodha Kite Connect (Instrument Token: **264969**) and tracked across 5 days to determine market fragility:

India VIX Reading	Regime Identifier	Tactical Risk Protocol
VIX < 12	LOW (Complacent)	Calm environment. Directional swing longs have highest statistical follow-through.
12 <= VIX <= 18	NORMAL (Healthy)	Standard trading conditions. Full position sizing and normal targets.
18 < VIX <= 25	ELEVATED (Cautious)	Heightened volatility. Tighten stops; favor Target 1 exits over multi-day runners.
25 < VIX <= 35	HIGH FEAR	Risk-off environment. Reduce position sizing by 50%.
VIX > 35	EXTREME PANIC	Market turbulence. Avoid all fresh swing entries; preserve cash.

12.2 Institutional Flow (FII / DII) & News Pipeline:

- **NSE FII/DII Net Flow:** Parses official NSE cash market participant data across 5 days (e.g. `STRONG_FII_BUYING > +Rs. 2,000 Cr`).
- **Google News RSS Pipeline:** Queries RSS feeds within 72 hours, strips syndicated noise, and surfaces the top 3 corporate catalysts.

13. Artificial Intelligence Synthesis & Strict Guardrails

AI ARCHITECTURAL SAFETY PROTOCOL

LLM Operational Boundary & Guardrails:

- 1. **Zero Mathematical Authority:** The Gemini 1.5 Flash model receives a complete JSON dictionary containing pre-computed indicators, states, and risk levels. It is *strictly prohibited* from altering or computing numbers.
- 2. **Institutional Prose Synthesis:** Translates the confluence findings, VIX state, FII flows, and news catalysts into a concise 4-6 sentence executive summary for the trader.
- 3. **Deterministic Fallback:** If the Gemini API key is missing or rate-limited, the system automatically activates a local rule-based template engine that compiles structured commentary with zero external API dependencies.

14. API Endpoints & Integration Contract

The FastAPI backend (running on <http://127.0.0.1:8000>) exposes the following REST interface:

Endpoint	HTTP	Payload / Query	Operational Function
/api/health	GET	None	Liveness probe and system status.
/api/auth/login-url	GET	None	Generates Zerodha Kite OAuth redirect URL.
/api/auth/callback	GET	?request_token=...	Exchanges request token for daily access session.
/api/auth/status	GET	None	Returns active user name ('Satyam Verma') and connection flag.
/api/analyze	POST	{"symbol": "TCS", "interval": "day"}	Runs complete 5-dimension deep-dive, risk matrix, and AI narrative.
/api/screener	POST	{"universe": "nifty100", "max_stocks": 100}	Scans universe, executes 3-tier bucketing, intra-bucket ranking, and breadth.
/api/signals	POST	{"short_sma": 6, "long_sma": 30}	Executes historical SMA crossover scanner.

Document Sign-off: Indian Equity Technical Advisory Agent Specification v2.0.0. All algorithms, mathematical formulas, and risk matrices are verified and operational against live Zerodha Kite Connect data feeds.